

AI Tool Usage

AI tools were used as supportive resources during this project mainly to improve coding efficiency and clarify statistical concepts. The primary tool used was Claude (Anthropic). All data cleaning, modeling decisions, statistical analysis, and interpretation of results were performed by our team members, and any AI suggestions were reviewed and verified before being used. Example prompts included questions such as “How can we parse nested dictionaries in pandas?” and “How can we run a chi-square test using a contingency table in Python?”

AI assistance helped structure the code for parsing and cleaning the Yelp attributes column, including flattening nested dictionaries such as Ambience and GoodForMeal and converting them into binary variables. AI also helped generate the function used to compute the average daily operating hours from the hours column. Our team wrote the function for counting days open and performed the final data cleaning and postal-code–based median imputation.

AI was also used to help implement the chi-square test step for attribute screening using a 2×2 contingency table (`pd.crosstab`) and to generate formatting code for the side-by-side bar chart comparing success rates for the top attributes. All statistical methods, feature engineering decisions, and visualizations were ultimately selected and verified by our team.